Projectile Motion Project - Honors

Experiment

Materials: ball bearing, launcher, meter stick, protractor, hoop, target.

Problem:

- Determine the launch velocity (v_0), the total time to fall (t), the max height (y_{max}) and horizontal displacement (x_f) of a projectile that is launched off of a balcony at some angle Θ.
- **Your objective is to place a target at the maximum horizontal displacement location on the floor. Your ball bearing should launch and land on the target.**

Task #1: Determine the launch velocity

- Design an experiment to determine the launch velocity of the projectile launcher (v_0)
- You must do this with only a meter stick. Stopwatches are NOT allowed.
- You should create an equation that allows you to solve for v_0 in terms of $x_f,\ y_0,\ and\ g$.
- This is the first thing you do *before* we launch anything off the balcony.

Task #2: Launching from the balcony

- You will then use v_0 to calculate the total time to fall (t), the max height (y_{max}) and horizontal displacement (x_f) for a projectile launched off a balcony at an angle.
- You will know v_0, y_0, x_0, g, and launch angle Θ (Jantz will give you the angle).
- You will need to derive equations using only variables that you know to determine the total time to fall (t), the max height (y_{max}) and horizontal displacement (x_f).
- You will then write a procedure that details exactly how you will experimentally determine these variables on launch day.

Pre-Lab Work: All work must be completed *prior* to utilizing any of the equipment

- All of the following work must be typed in your lab document.
    - Brief description of background information on projectiles. This includes the assumptions you are making that provide the framework for the experiment.
    - Complete procedure for determining the launch velocity.
    - Diagram of experimental setup utlized.
    - Theoretical equation that allows you to solve for v_0 in terms of $x_f,\ y_0,\ and\ g$.
    - Description of anticipated sources of error and what you will do to minimize error.

Submission

- Formal Lab Write-Up
- One submission per partner group

Rubric

Points	Item	Description
5	Title / Objective	Description of what was observed and how the investigation was carried out.
20	Stated Theory	Includes background information about projectiles. You should talk about the assumptions that you are making, the setup that you are using, etc. The theory here is the information that provides the framework that you approach the experiment through.
20	Equations	Use your theory to derive equations for: ● Experiment #1: v_0 in terms of x_f, y_0, $and\ g$. ● Experiment #2: Explanation along with your equations for how you will determine time, max height, and horizontal displacement. You should explain what each variable means, what the constants are, and the assumptions that you are making. This section can be combined with the theory section to make the report flow. ● Define variables and constants. ● Derive and/or explain the relationship between variables and constants. ● Show how these equations are derived from your basic kinematic equations.
20	Experimental Design	**Setup:** Qualitatively describe the setup that will assist in precise measurements for each experiment. Diagram or drawing of the relevant components of the apparatus, accompanied with description. Each setup and procedure should be included. **Procedure:** Describe the procedures performed including quantities that are held constant, varied and measured. Include theoretical equation. Be sure to include how you plan to determine error in measurement. **The setup and procedure must be done for both experiment #1 and #2.**
10	Data Table	**Data Table**: Should be included in an appendix at the end of the report. You should include ALL numbers that you measure and collect throughout the report. Present this in a clean, easy to read fashion.
15	Data Analysis	What is the significance of the results and how did the experimental data compare to the predicted data? Statements should include qualitative and quantitative analysis. ● You should talk about the limitations of the measuring tools. Be specific! ● You should calculate a percent error between your expected values and your measured values.
10	Improvements and Extensions	Short statement of how the investigation could be improved to reduce error. If error is minimal (<10%) suggest next step to extend knowledge of the subject.